25901109

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBTG200, MG4DSCBBT201 - MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

SCHEME

Part A-One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. Name the purine bases present in DNA. [K] / [CO1]

   Adenine and Guanine- 2 marks

2. Who provided definitive evidence that DNA is the genetic material of bacteriophages? [K] / [CO1]

   A D Hershey and Martha Chase – 2 marks

3. Identify the noncoding region of a gene. [U] / [CO2]

   Introns- 2 marks

4. Name any 3 arms of tRNA. [K] / [CO3]

   Any three arms- 2 marks

5. Name the protein that initiates DNA replication. [K] / [CO3]

   DnaA protein/Helicase/ORC complex- 2marks

6. Which radiation commonly causes thymine dimers? [K] / [CO4]

UV radiation- 2 marks

7. What is the function of the poly-A tail? [K] / [CO3]

   Protect mRNA from degradation- 2 marks

8. What enzyme degrades RNA in RNA-DNA hybrid? [K] / [CO3]

   Ribonuclease H- 2 marks

9. Who proposed Wobble Hypothesis? [K] / [CO3]

   Watson and Crick- 2 marks

10. Which RNA is modified? [K] / [CO3]

    HnRNA/Primary RNA/rRNA/tRNA/mRNA- 2 marks

11. Recognize the lac operon as an inducible or repressible operon. [K] / [CO4]

    Inducible- 2 marks

12. Name the DNA sequence to which a repressor binds. [K] / [CO4]

    Operator- 2 marks

[2x10 = 20]

Part B-Short Essay-Answer any 5 questions

Each question carries 4 marks

13. Explain X-Ray Crystallographic Studies conducted by Rosalind Franklin. [K] / [CO1]

    Role of X-Ray Crystallographic Studies in DNA structure identification- 4 points- 4 marks

14. Explain the role of hydrogen bonding and base stacking in DNA stability. [U] / [CO1]

    Hydrogen bonding- double bond, triple bond-role in DNA stability- 3 marks, base stacking- 1 mark.

15. Explain the structure of a nucleosome. [U] / [CO2]

    Structure- 3 marks, diagram- 1 mark

16. Explain why DNA polymerase requires a primer. [U] / [CO3]

    DNA polymerase- minimum 3 points- 3 marks, primer- 1 mark.

17. Explain the role of the Shine– Dalgarno sequence in initiation. [U] / [CO3]

    Shine– Dalgarno sequence- definition- 2 marks, role in initiation- 2 marks.

18. Explain the role of eukaryotic initiation factors (eIFs) in translation. [U] / [CO3]

    4 eukaryotic initiation factors- 2 marks, role- 2 marks

19. Explain copy-paste transposons. [K] / [CO4]

Definition- 2 marks, example- 2 marks. [4x5 = 20]

Part C-Essay type questions-Answer any 1 questions

Each question carries 10 marks

20. Explain the mechanism of DNA replication in Prokaryotes with a neat labelled diagram. [U] / [CO3]

    Introduction- 2 marks, Mechanism- 4 marks, labelled diagram- 4 marks.

21. Explain how attenuation controls trp operon. [U] / [CO4]

    trp operon.-introduction- 2 marks, structure- explanation- 5 marks, Attenuation-2 marks, diagram- 1 mark

[10x1 = 10]